Project

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1. Project

Faculty:	Information Technology
Module Code:	ITNSA2-11
Module Name:	Network Security Block A
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Total Marks:	100
Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is a group project.

Groups should consist of minimum of three (3) members.

This project contributes 50% towards the final mark.

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

2. AI Checklist and Declaration

Before you submit an assignment, you should be able to confidently and honestly make all the below statements. For group work, you can also review the list, together, to hold one another accountable.

- I confirm that my submission reflects my personal learning, knowledge, skills, and understanding.
- If AI tools were employed for generating any part of this assignment (even in the drafting/research phase), I have referenced the use of AI in the text and/or declared the use of AI. I am willing to discuss the process and its contribution to my learning.
- I am aware that the lecturer may request a demonstration of my learning, such as explaining choices in approach, research, and the content I am submitting.
- I am aware that, if I did use AI in any phase of preparing this submitted work, it is recommended that I save a copy of the relevant chat history (prompts and answers), as this will help me demonstrate my writing/work process to my lecturer, if I am asked to do so.
- I have read the assignment instructions on whether AI tools are prohibited for this assignment, and if they are prohibited, I can confirm that I did not use AI tools.
- I understand that failure to agree to these terms may be deemed unethical, potentially leading to disciplinary action. I understand my responsibility for the integrity of my work, including seeking clarification from academic staff and adhering to instructions.

It is essential to acknowledge your use of ChatGPT or other generative AI in your learning. If you use ChatGPT or other generative AI to help you generate ideas or plan your process, you should still acknowledge how you used the tool, even if you don't include any AI-generated content in the assignment.

Please note: The following guiding questions that you will be asked in an AI declaration questionnaire below this assignment brief.

AI Declaration

It is compulsory to complete this AI declaration for each of your assignment submissions.

I carefully read the assignment instructions, and the extent to which AI may be used for the assignment.
I used the following AI system(s)/tool(s):
I used it for the following:

If I quoted or paraphrased an AI output, I have referenced the relevant tool, version, and the date I used the tool.
I still consider this work my own. (i.e., I have not outsourced the final product, or significant portions of it, to AI tools/systems).
If required, I can defend my argument/perspective, explain my choices and approach, and can show that I am knowledgeable about the details of my work.

For further guidance on the use of AI at Eduvos, please refer to the AI FAQ glossary. You will locate the FAQs in the Artificial Intelligence tile on the myDocuments page of myLMS.

3. Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted assignments.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your assignments.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.
7. One group member should be nominated to submit the assessment on behalf of the group. Multiple submissions by various group members will result in an inflated similarity index on Turnitin.

4. Section A

Section A

Learning Objective

Demonstrate an understanding of network security fundamentals and how they are applicable in real-life scenarios by creating virtual environments that would mimic what real world organisations would have. In addition to understanding networking principles, a solid knowledge of how Cisco Packet Tracer works is essential.

Project Topic

Network Security, network design, network configuration.

Scope

Week 1 - 7

Technical Aspects

Creating simulated virtual networks

Marking Criteria

Verify All Configurations in Cisco Packet Tracer

4.1. Question 1

Question 1 35 Marks

Study the scenario and complete the question(s) that follow(s):

BusyBugs, a web application development company based in Cape Town, occupies three floors to accommodate its growing teams: **Development**, **Product Owners**, **Testers**, and **UX Developers**. Due to rapid expansion and the introduction of a new **staging environment** for client demonstrations, the company's network requirements have become more complex, demanding stricter segregation and controlled external access. Here is the revised scenario:

1. **Floors and Teams**

   - **Floor 1**: Development (3 end user devices) and Product Owners (3 end user devices), each with its own DHCP server.
   - **Floor 2**: Testers (2 end user devices) and UX Developers (2 end user devices). These teams physically share the same floor but must be **logically separated** into distinct VLANs for security.
   - **Floor 3**: Newly introduced **Staging Environment**, consisting of a test web server used to demonstrate partially completed features to select external partners. This staging server must be isolated from the rest of the internal network except for essential development traffic.

2. **DHCP and DNS**

   - Each continue to manage their own DHCP servers.
   - The **DNS Server** now needs to handle both internal name resolution (for local devices on all floors) and external resolutions (for staging demos and general internet access).

3. **Routers and Routing**

   - **Product Owners' Router**: Connects Floor 1 to the DNS server, web servers, and the internet.
   - **Developers' Router**: Connects Floor 1 to the DNS server, staging environment, and the internet.
   - **Shared Router**: Used by **Floor 2** (UX Developers, Testers) **and** the new staging environment on Floor 3, to reduce costs.
   - **Routing Protocols**:
   - RIP remains enabled on one router for backward compatibility with certain legacy systems.
   - **OSPF** is introduced on the Shared Router for scalability and efficient routing to the staging environment.

4. **VLANs**

   - The **UX Developers** and **Testers** must be on separate VLANs, even though they share Floor 2. This requires **802.1Q trunking** and appropriate **inter-VLAN routing** (on a router or a Layer 3 switch).
   - The **Staging Environment** on Floor 3 is placed in its own VLAN, partially accessible from the internet for client demos. Traffic to this VLAN must be **strictly filtered** to prevent unauthorised access to internal resources.

5. **Servers**

   - **Web Server**: Hosts the main production applications. Accessible internally by Development, Product Owners, UX Developers, and Testers.
   - **Staging Web Server**: Located on Floor 3, used for demos to select external partners and accessible to the Development team for updates.
   - **DNS Server**: Provides local name resolution for all floors, plus external DNS lookups.

6. **Security & Access Control**

   - **DMZ-like Access**: The staging web server requires inbound connections from approved external partner IPs only, with restricted ports (e.g., HTTP/HTTPS).
   - **ACLs**: Implement extended ACLs to limit which internal VLANs/Subnets can reach the staging server and to restrict staging access to the production web server.
   - **NAT or Port Forwarding**: Configure NAT so external users can access the staging server using a public IP address without exposing other internal services.
   - **VLAN Separation**: UX Developers and Testers must not access each other's VLAN, except where minimal testing collaboration is authorised (e.g., one shared file server or printer).

- **Monitoring**: A plan for **log collection** on each router (including RIP and OSPF events), plus basic intrusion detection to track unusual traffic patterns around the staging VLAN.

7. **Additional Challenges**

- **Subnet Planning**: Allocate IP ranges for each VLAN such that they accommodate both existing and future device counts.
- **Redundancy**: Consider a second link between the Shared Router and the Product Owners' Router for failover, requiring proper route advertisement in both RIP and OSPF.
- **DHCP Reliability**: Ensure each DHCP server can handle conflicts or failovers, so if one fails, devices on that floor can still obtain IP addresses.
- **Performance**: Use trunking and inter-VLAN routing optimisations (e.g., EtherChannel on core switches, if applicable) to handle increasing traffic without bottlenecks.

1.1 **Create the Base Topology**

(20 Marks)

- In Cisco Packet Tracer, design a network layout reflecting the three floors:
  - Floor 1: Development (3 devices) + Product Owners (3 devices).
  - Floor 2: Testers (2 devices) + UX Developers (2 devices).
  - Floor 3: Staging Environment (1 staging web server).
- Include the routers as per the scenario:
  - **Product Owners' Router** (Floor 1)
  - **Developers' Router** (Floor 1)
  - **Shared Router** for Floor 2 and Floor 3.

1.2 Configure separate DHCP servers for the Development and Product Owners teams on Floor.

(10 Marks)

1.3 Install/configure a DNS server reachable by all workstations.

(5 Marks)

End of Question 1

4.2. Question 2

Question 2 20 Marks

2.1 Implement two VLANs: one for Testers and another for UX Developers.

(10 Marks)

2.2 Configure the Shared Router (or a Layer 3 switch) to enable traffic between VLANs.

(5 Marks)

2.3 Implement an ACL (or equivalent mechanism) to block direct communication between the Testers VLAN and the UX Developers VLAN, unless explicitly required.

(5 Marks)

End of Question 2

4.3. Question 3

Question 3 25 Marks

3.1 Using Cisco Packet Tracer, integrate the staging environment on Floor 3 such that external partners can reach its web service via HTTP/HTTPS while maintaining restricted access to the rest of the network. Demonstrate NAT/port forwarding, ACL rules, and Development team update permissions, providing screenshots or command outputs that show how you balance security with the need for external demos.

(15 Marks)

3.2 In addition to implementing the staging environment in Packet Tracer, discuss the theoretical design of how you would integrate this partially externally accessible environment into BusyBugs' network.

(10 Marks)

End of Question 3

4.4. Question 4

Question 4 20 Marks

4.1 In Cisco Packet Tracer, enhance network resilience and monitoring by enabling RIP on one router for legacy compatibility and OSPF on the Shared Router for efficient routing. Show how you would introduce a failover or backup link to maintain connectivity if a primary link fails.

(10 Marks)

4.2 Discuss the challenges of using both RIP and OSPF in the same network, particularly when introducing a failover or backup link to ensure resilience.

(10 Marks)

End of Question 4